

KEFI AI: MASTER TECHNICAL PROJECT REPORT

AI-Powered Dynamic Mental Health Monitoring & Distress Prediction System
for Victims of Atrocities
Smart India Hackathon (SIH) 2026 — Problem Statement ID: SIH26094

1. Official Team Details (Team DEKO)

- Team Name: DEKO
- Team ID: 141444
- Team Leader: BALAJI P (balajikrishnan031@gmail.com | 9342636595)
- Institute: University College of Engineering, Panruti
- Ministry / Problem Creator: Ministry of Social Justice and Empowerment (MoSJE) / MIC

Role	Member Name	Email	Phone	Gender
LEADER	BALAJI P	balajikrishnan031@gmail.com	9342636595	Male
TEAM_MEMBER	NAVANEETHAM V	er.navaneetham@gmail.com	9487461627	Female
TEAM_MEMBER	MALINI V	malini28102005v@gmail.com	8807984385	Female
TEAM_MEMBER	DHAVAN R G	rgdhavan50@gmail.com	9342529181	Male
TEAM_MEMBER	DEEBIKA S	deebikasivaprakasam@gmail.com	6380433293	Female
TEAM_MEMBER	DHARA R	dhararamesh2416@gmail.com	9043644162	Female

2. Executive Summary & Solution Concept

Victims and witnesses of atrocities registered under the Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act, 1989 frequently experience severe, unaddressed psychological distress following complaint registration. Stress escalates due to threats from accused persons, repeated court appearances, investigation delays, social ostracism, economic hardship, and rehabilitation hurdles. Existing mechanisms focus on post-hoc monetary relief and legal aid, leaving a major void in continuous mental health monitoring.

KEFI AI (Kinematic Emotion & Forecasting Intervention System) is created specifically for MoSJE to bridge this void. Derived from Greek Kéfi (Κέφι) — meaning inner resilience, emotional vitality, and healing after trauma — KEFI proactively contacts victims through IVRS 14566, Chatbot, SMS, Mobile App, and Web Portal, analyzing text, voice acoustic prosody, behavior, and threat keywords.

3. Mathematical Dynamic Distress Score (DDS) & Forecasting Engine

KEFI AI computes a continuous Dynamic Distress Score (DDS: 0–100) using the weighted formula:

$$\text{DDS} = 0.25 * E + 0.20 * V + 0.15 * B + 0.15 * H + 0.10 * C + 0.15 * T$$

- E (0.25) — NLP Sentiment & Emotion AI Score (Fear, Hopelessness, Sadness valence)
- V (0.20) — Vocal Acoustic Prosody Stress Index (Pitch tremor Hz, Jitter %, Pause gap duration)
- B (0.15) — Behavioral Check-in Gap Rate (Missed periodic wellness calls)
- H (0.15) — Historical Trend Velocity (dDDS/dt over past 7 days)
- C (0.10) — Case Stage Pressure (Court hearing proximity & trial delays)
- T (0.15) — Threat Keyword Density ('dhamki', 'withdraw', 'court')

72-Hour Longitudinal Crisis Forecasting: Uses time-series velocity (dDDS/dt) to predict psychological crises 72 hours before escalation occurs.

4. 3-Tier Risk Classification & Action Matrix

1.  Green / Yellow Band (0–30 Score): Routine support, daily wellness check-ins, self-help box breathing, 5-4-3-2-1 sensory grounding audio, SC/ST PoA Act legal rights pamphlets.
2.  Orange Band (31–60 Score): Focused check-ins, scheduled tele-counselling follow-up, legal aid consultation.
3.  Red Band (61–100 Score): Real-time alert dispatched to District Magistrate (DM) / SP dashboard, witness protection unit, and emergency crisis intervention team.

5. Explainable AI (SHAP/LIME XAI) & DPDP Act 2023 Compliance

To prevent 'black box' AI decisions, KEFI outputs exact point attributions (+21 threat language, +18 fear sentiment, +14 trend velocity, +12 missed check-ins, +9 voice jitter). Fully DPDP Act 2023 compliant with end-to-end AES-256 encryption, OTP digital consent, and victim data purge controls.

6. Core Platform Features & User Experience

- Granular Plutchik Emotion Wheel UI (Scared/Intimidated, Sad/Isolated, Angry/Resentful, Anxious/Overwhelmed, Exhausted/Burnout, Calm/Hopeful)
- Calm Space & Silent Sanctuary (Box Breathing 4-4-4-4, 5-4-3-2-1 Grounding, Ambient Audio soundscapes)
- Silent Mode & Body Language Detection (Camera/wearable posture & face affect tracking for

users who cannot speak/type)

- Progress Dashboard & Weekly Report Card (7-day trend graph, 12/14 micro-goals achieved, )
Low Risk)
- Tanglish & Multilingual Conversational AI (Supports Tanglish தமிழ்-English, Tamil, Hindi, English)
- Emergency Quick SOS Button (Instant dispatch to 14566 Atrocity Helpline & Police Liaison)